Julianna Russo, Gianna Julio, Daisy Kopycienski CMPT417 - Hacking and Pen Testing
02/02/2024 Image Forensics

This assignment investigates the metadata attached to an image and steganography, the concealing of information within an image.

To find this information, we downloaded the image jpg file and then uploaded it to the online EXIF tool here: https://exif.tools/. This provided us with the additional data used to answer questions 1-3. It also gave us resolution data, image size info, and coloring/editing data.

1) What type of camera was used to take this picture (manufacturer and model number)?

Motorola XT907

2) When was this photo taken (include date, month, year, hour, minute, and second)?

2013:04:27 12:59:24

3) Did I use a flash when I took this picture?

No Flash

For question 4, we knew there was a hidden string of text within this image file. First, we attempted to decode it using online steganography tools, but they were unable to detect the flag. Then, we opened the jpg file with a basic text editor application and could search for the hidden string.

4) Where is the flag (easyctf{xxx}) hidden in the apple?

the flag is easyctf{w0w_much_appl3s}

To find this, we opened the jpg file as a text file with the TextEdit application and searched for the flag keyword: ‘easyctf.’ The message is hidden within the image data in plaintext.

apple.jpg

easyctf

Done Replace

the flag is easyctf{w0w_much_appl3s}